

INDUSTRIAL PRODUCT DATASHEET

RTL8852BE Wireless NIC, 2.4G/5GHz Dual-Band WiFi 6, Bluetooth 5.2, Compatible with Jetson Orin

The RTL8852BE Wireless NIC delivers 2.4G/5GHz dual-band Wi-Fi 6 at up to 1201Mbps with Bluetooth 5.2 in M.2 Type 2230 for Jetson Orin, edge AI boxes, robotics, industrial IoT, and consumer electronics.

SKU

100042928

UPDATED

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PRICE

\$13.50(Excl. VAT)



Overview

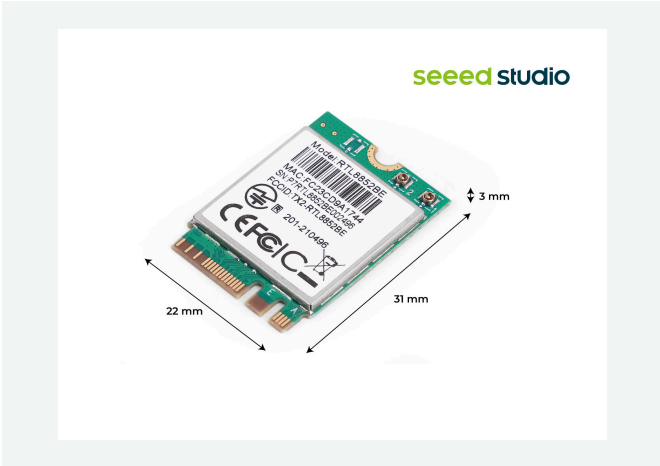
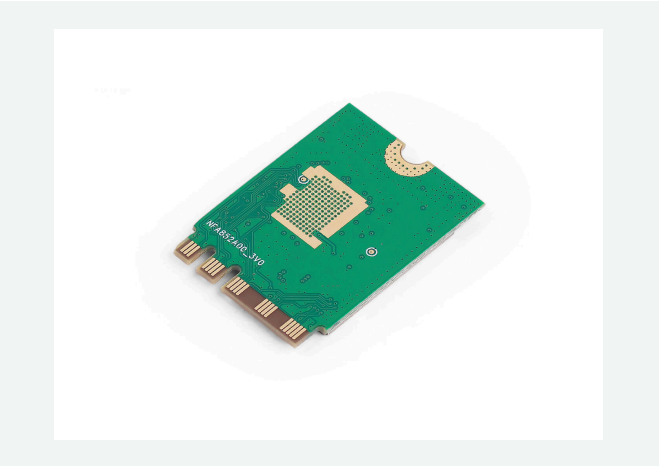
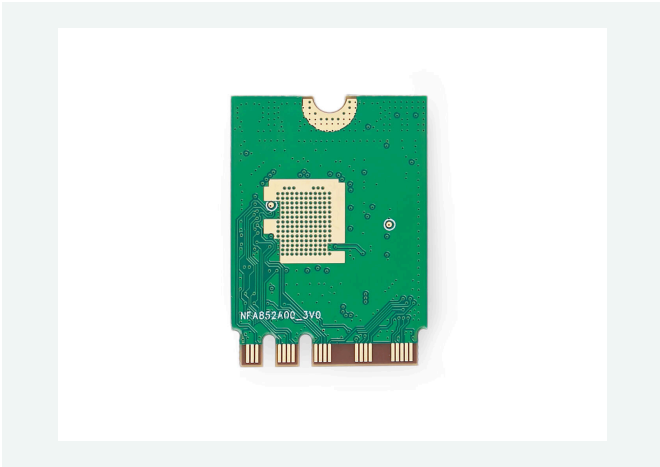


The RTL8852BE Wireless NIC is a 2.4G/5GHz dual-band Wi-Fi 6 and Bluetooth 5.2 module in an M.2 Type 2230 S2 Key A+E form factor, delivering PHY rates up to 1201Mbps and dual-mode Bluetooth (LE/BR/EDR). It is compatible with Jetson Orin and widely used in Edge AI boxes and Mini PCs, robotics and embodied AI, industrial IoT, and consumer electronics.

Key Features

- Supports 2.4G/5GHz dual-band
- Wireless PHY rate can reach up to 1201Mbps
- IEEE Standards: IEEE 802.11a/b/g/n/ac/ax
- Dual Mode Bluetooth support : Simultaneous LE and BR / EDR
- Form factor: M.2 Type 2230 S2 Key A+E Card
- Connect to external antenna through MHF4/IPEX4 connectors

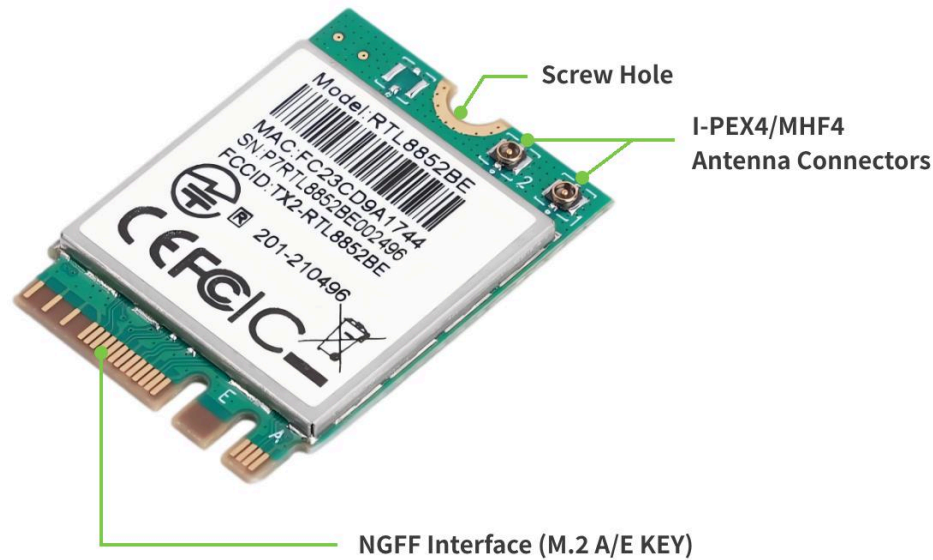
Product Gallery



Specification

Chipset	RTL8852BE - CG
WLAN Standards	IEEE 802.11a/b/g/n/ac/ax
BT Specification	Bluetooth Core Specification v5.2/4.2/2.1
Host Interface	PCI Express 1.1 for WLAN & USB2.0 FS for Bluetooth
Antenna	Connect to the external antennas through MHF4 connector (Two antennas)
Dimension	M.2 Type 2230 Pluggable Card: 30*22*2.15mm (L*W*H)
Power Supply	DC 3.3V±0.2V @ 1.5A (Max)
Operation Temperature	-20°C to +70°C
Operation Humidity	10% to 95% RH (Non - Condensing)

Hardware Overview



Applications

- **Edge AI Box & Mini PC:** Supports Linux and Windows. Featuring a universal interface, it is widely integrated into various Edge AI Box.
- **Robotics & Embodied AI:** Provides the low-latency, high-reliability wireless link and Bluetooth 5.2 connectivity essential for commercial service robot and AGV/AMR.
- **Industrial IoT:** Ideal for harsh industrial environments where wired networking is impractical. It leverages Wi-Fi 6 for high-capacity data collection and transmission, with wide-temperature support making it highly popular in industrial application.
- **Consumer Electronics:** Frequently used as a Wi-Fi 6 upgrade from legacy Wi-Fi 5 cards, significantly boosting the wireless networking performance of Laptop and PC.

Part List

Wireless RTL8852BE-CG	x1
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Compliance & Logistics



HSCODE	8517799000
USHSCODE	8517690000
UPC	
EUHSCODE	8517180000
COO	CHINA

FCC Requirement

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) this device may not cause harmful interference, and
- (2) this device must accept any interference received, including interference that may cause undesired operation.

15.21

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates, uses, and can radiate radio frequency energy.

If not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Resource Links

- <https://www.seeedstudio.com/RTL8852BE-WIFI-Module-M-2-Key-E-p-6835.html>
- [Datasheet](#)

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